Computer Network Security and Risk Management

Introduction

Computer networks allow devices to exchange information and access shared resources. Organisations use networks to support communication, cloud services, financial transactions, customer management, and operational systems. As dependence on digital infrastructure increases, protecting networks becomes essential for maintaining confidentiality, integrity, and availability.

Network security involves technologies, policies, and procedures designed to prevent unauthorised access and reduce the impact of cyber incidents. Threats may originate from malicious software, compromised accounts, vulnerable devices, misconfigured systems, or human error. Effective protection requires both technical safeguards and responsible organisational practices.

Common Network Threats

Malware is software designed to perform harmful or unauthorised actions. Different forms include ransomware, spyware, worms, and trojans. Ransomware may encrypt files and demand payment, while spyware can collect information without the user's knowledge. Worms can spread between vulnerable systems, potentially disrupting services.

Phishing attacks attempt to persuade people to reveal credentials, open malicious attachments, or visit fraudulent websites. Attackers may imitate trusted organisations or create messages that appear urgent. Training can help users recognise suspicious requests, but technical controls are also needed because deceptive messages can be difficult to identify.

Denial-of-service attacks attempt to make a service unavailable by overwhelming its resources or exploiting weaknesses. Distributed attacks may use many compromised devices to generate traffic simultaneously. Organisations can reduce their impact through traffic filtering, capacity planning, monitoring, and appropriate incident-response arrangements.

Authentication and Access Control

Authentication verifies the identity of a user or device before access is granted. Passwords remain common, but weak or reused passwords can expose accounts to attacks. Multi-factor authentication adds another verification step, making unauthorised access more difficult when a password is compromised.

Access control determines which resources an authenticated user may use. The principle of least privilege recommends granting only the permissions required for a

person's responsibilities. Administrators should review access rights regularly, particularly when employees change roles or leave an organisation.

Role-based access control can simplify permission management by assigning privileges to job functions. Sensitive administrative accounts may require additional restrictions and stronger authentication. Audit records can help investigators determine which accounts accessed particular resources and when those activities occurred.

Encryption and Secure Communication

Encryption transforms readable information into a protected representation that can be recovered with the appropriate key. It is used to protect data transmitted across networks and data stored on devices. Secure communication protocols help reduce the risk that attackers can read or modify information in transit.

Public-key cryptography uses related public and private keys for operations such as encryption and digital signatures. Symmetric encryption uses a shared secret key and is often efficient for protecting large amounts of data. Secure systems must manage keys carefully because the protection offered by encryption depends on keeping sensitive keys safe.

Digital certificates can help verify the identity of systems involved in secure connections. Organisations should maintain certificates, update cryptographic configurations, and avoid obsolete protocols. Encryption is not a complete security solution, since compromised endpoints and stolen credentials can still expose protected information.

Security Monitoring and Incident Response

Network monitoring collects information about traffic, system events, authentication attempts, and unusual activity. Security teams analyse these records to identify potential threats and investigate incidents. Automated alerts can help prioritise suspicious events, although poorly configured rules may generate large numbers of false alarms.

An incident-response plan defines how an organisation detects, contains, investigates, and recovers from security incidents. Initial actions may include isolating affected systems, preserving evidence, disabling compromised credentials, and informing relevant personnel. Recovery should include verification that systems are safe before they return to normal operation.

After an incident, organisations should examine its causes and identify improvements. A review may reveal missing patches, excessive privileges, insufficient monitoring, or weaknesses in staff training. Applying these lessons can reduce the likelihood or impact of similar incidents in the future.

Risk Assessment and Security Governance

Risk assessment evaluates potential threats, system vulnerabilities, and the consequences of successful attacks. Organisations can use these findings to prioritise security investments. Critical systems may require stronger safeguards because their failure could disrupt essential operations or expose sensitive information.

Security policies establish expectations for password management, device use, software installation, data handling, and incident reporting. Policies should be supported by practical procedures and periodic reviews. Employees need to understand their responsibilities and know how to report suspicious events.

Backups are an important part of resilience planning. Protected backups can help restore information after accidental deletion, hardware failure, or ransomware incidents. Organisations should test restoration procedures and ensure that backup copies cannot easily be altered by an attacker who compromises the main environment.

Conclusion

Computer network security requires a combination of authentication, access control, encryption, monitoring, incident response, and risk management. No single technology eliminates every threat. Organisations must regularly evaluate their systems, apply appropriate updates, protect sensitive information, and train personnel. A coordinated security programme can reduce exposure to cyber threats and improve the resilience of digital services.